

Hyperliquid Ranking

Unleashing the meta model on chain



CROWDCENT

Robert Smillie



@rsmillie94



rsmillie94



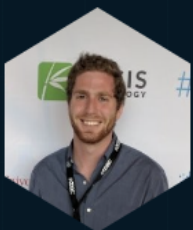
CROWDCENT

- The Numerai meta model has achieved a 25% return over the past year with an average correlation of 2.9%.
- In the crypto tournament the average correlation over the past year has been 16.6%.
- Can we make 6x the return with 6x the correlation?
- Let's figure out how!

About CrowdCent

CrowdCent is on a mission to decentralize investment management by changing the way investment funds make decisions and allocate capital. We are the machine learning and coordination layer for online investment communities looking to turn their data into actionable, investable portfolios.

By combining human insight with artificial intelligence, we're creating a new type of investing where investment ideas and algorithms are judged purely on merit, not pedigree. We enable fundamental analysts and data scientists worldwide to collaborate and compete based on their contributions — creating investment strategies that crowdsource the best of both human and machine intelligence.



Jason Rosenfeld

CO-FOUNDER & CEO

Jason is a "full-stack" machine learning engineer and value investor. Before founding CrowdCent, he worked as a Data Scientist & Investment Analyst at Millennium Management and as an Equity Research Associate at Canaccord Genuity. Jason is a [Scikit-learn Expert](#) and Numerai Grandmaster. He has a Master's in Information and Data Science from UC Berkeley with a focus in Natural Language Processing and a Bachelor's in Mathematical Economics from Colgate University.



Ryan Mahon

CO-FOUNDER

Ryan is an investor, technologist, and outdoor enthusiast. Ryan most recently ran the Mobility Strategy at Ibex Investors. Previously, he worked as a Sector Head (Internet) and Associate Portfolio Manager at Millennium Management and as an Investment Analyst (Global Technology) at Folger Hill Asset Management. He graduated magna cum laude from Georgetown University, McDonough School of Business, with a BSBA in Finance, Accounting, and Psychology, and is a CFA Charterholder.

Who am I?

- I am an actuary from the UK who specialises in optimisation, process improvement and AI.
- Numerai Grandmaster.
- Not officially affiliated with CrowdCent.
- Love the idea, have been helping out with user testing and wanted to present to bring some light to the platform.

Why build a different competition

- Inspired by Numerai, not replacing it.
 - Numerai's crypto tournament is thriving, and it is proof that collective intelligence works in digital assets too.
- Market neutral requires long + short.
 - To capture pure alpha, you can't just predict winners — you need to predict both winners and losers.
 - Numerai follows this ethos with their market neutral hedge fund. This is even more important in crypto with unpredictable cycles.
 - We needed a prediction universe where long and short positions are directly tradeable.

- The problem:
 - The Numerai universe contains many assets which are basically impossible to trade in both long and short directions.
- Our goal:
 - Build a framework where models could be directly mapped into tradable strategies in a market-neutral setting.
- Hyperliquid makes it possible.
 - With ~170 perpetual markets, each tradeable long or short with leverage, Hyperliquid provides the perfect on-chain playground for market-neutral crypto strategies.

- The result:
 - The CrowdCent Hyperliquid Ranking competition is a crypto-native complement to Numerai — built to test predictive intelligence in a tradable, long/short, market-neutral universe of hyperliquid assets.

Challenges / Hyperliquid Ranking

Hyperliquid Ranking

Runs: Jun 05, 2025 - Ongoing

Description

Build machine learning models to predict the future, relative price movement of crypto assets listed on Hyperliquid 10 days and 30 days from now.

We are not affiliated with Hyperliquid.

PARTICIPANTS

15

SUBMISSIONS

2270

PRIMARY METRIC

composite_percentile ⓘ ↗

ⓘ Higher is better

Last 30 days

Last 30 days

Hyperliquid: Origins

- Founded by Jeff Yan (2023): Launched as a self-funded project after the founder's success in high-frequency crypto trading.
- Core team of ~11 people; no venture capital to maintain user-centric focus.
- Mission: Build a decentralized perpetuals exchange with the speed and experience of a centralized exchange, but with users in full control of funds.
- Inspired by the FTX collapse to create a trustless alternative for derivatives trading.
- Growth: Saw rapid adoption – processed ~\$330B in trading volume in July 2025, briefly surpassing Robinhood's volume.
- Through community-focused strategies (e.g. airdrops, wallet integrations), Hyperliquid has become one of DeFi's top derivatives platforms in just a year.

Hyperliquid: Technology & Key Features

- Custom Layer-1 (HyperCore + HyperEVM):
 - CEX-like speed: ~200k tx/sec, ~0.2s latency
 - HotStuff-style PoS consensus (HyperBFT) → fast, ordered finality
- On-Chain Orderbook (CLOB):
 - Every order & trade on-chain
 - Transparent + efficient (no AMMs, no off-chain servers)

Hyperliquid: Technology & Key Features

- CEX-Grade Trading:
 - Perpetuals with leverage, cross-margin, advanced order types
 - Low fees, instant settlement, deep liquidity
- Vaults for Liquidity:
 - HLP vault = market-making & liquidation engine
 - User Vaults for custom strategies, copy-trading, algos
- Differentiator:
 - All matching & margining on-chain without speed trade-off
 - ~75–80% share of decentralized perpetuals market

CrowdCent Hyperliquid Competition – Overview

- **Inspired by Numerai, built for crypto**
 - Same ethos: crowdsourced ML models aggregated into a meta-model.
 - Different universe: ~170 tradable crypto assets instead of equities.
- **What it is:**
 - Ongoing ML challenge to **rank tokens by 10-day & 30-day forward returns**.
- **Why it's unique:**
 - Signals translate directly into **long/short crypto portfolios**.
 - **Live DeFi markets** → predictions face real, evolving volatility.
 - Pure focus on **market-neutral alpha**.
- **Endgame:** Crowd intelligence → **MetaModel** → investable crypto strategies.

How it Works

Continuous format: Daily data drops (~14:00 UTC) → ~4h to submit predictions → rolling evaluation & live leaderboard.

Data: Starter training set + daily inference features (momentum, sentiment, tech indicators). External data allowed.

Targets: Predict **relative return ranks** over 10d & 30d horizons.

- Scores from 0–1 = rank ordering of assets.
- Designed for **long/short portfolio construction**.

Getting Started:

- **Tutorial notebook** is available here:
<https://docs.crowdcent.com/tutorials/hyperliquid-end-to-end/>
- Trains an XGBoost model on the training data and submits into one of your prediction slots.

Scoring and Metrics

- NDCG@40 (Symmetric):
 - Focus on top 40 & bottom 40 assets.
 - Rewards identifying big winners (long) & losers (short).
 - Perfect = 1.0; random $\approx$ 0.55.
- Spearman Correlation (ρ):
 - Measures overall rank alignment with true outcomes.
 - Complements NDCG by valuing consistency across the full list.
- Composite Score:
 - Combines both metrics (10d & 30d) into a percentile leaderboard score.
 - Higher percentile = stronger, more consistent model.
- Fairness Controls:
 - Predictions normalized before scoring.
 - Missing assets default to neutral rank (0.5).

rsmillie94's Scores

USER (AVERAGE)

SUBMISSION SLOTS

Performance Charts - Hyperliquid Ranking

PER PERIOD

CUMULATIVE

spearman_10d

Range: -0.314 - 0.508

μ : 0.032

σ : 0.171



spearman_30d

Range: -0.134 - 0.334

μ : 0.083

σ : 0.116



composite_percentile

Range: 13.710 - 100.000

μ : 55.647

σ : 20.940



ndcg@40_10d

Range: 0.444 - 0.772

μ : 0.569

σ : 0.068



ndcg@40_30d

Range: 0.474 - 0.700

μ : 0.592

σ : 0.054



Tutorial Highlight: LSTM vs XGBoost on CrowdCent Data

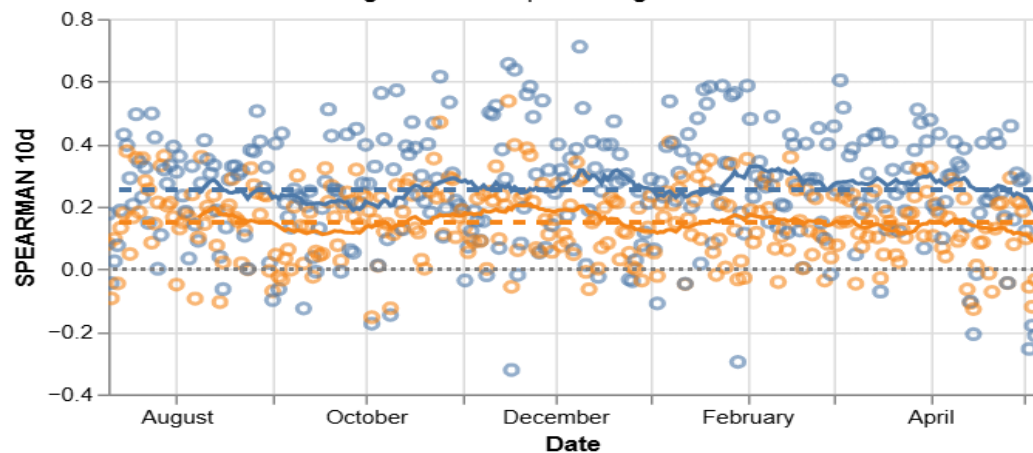
- Context:
 - CrowdCent's features are temporal (lag windows) → ideal for sequence models.
 - Traditional models (e.g. XGBoost) treat lags as flat inputs, missing time dependencies.
- Key Insight:
 - LSTM processes lagged features sequentially → captures temporal structure.
 - Reshapes inputs into time steps × features (instead of a flat vector).
- Results:
 - XGBoost: ~0.06 avg Spearman (30d)
 - LSTM: ~0.19 avg Spearman (30d) → 3x improvement
 - Gains consistent across metrics (Spearman & NDCG).
 - 10d horizons score higher than 30d.

Tutorial Highlight: LSTM vs XGBoost on CrowdCent Data

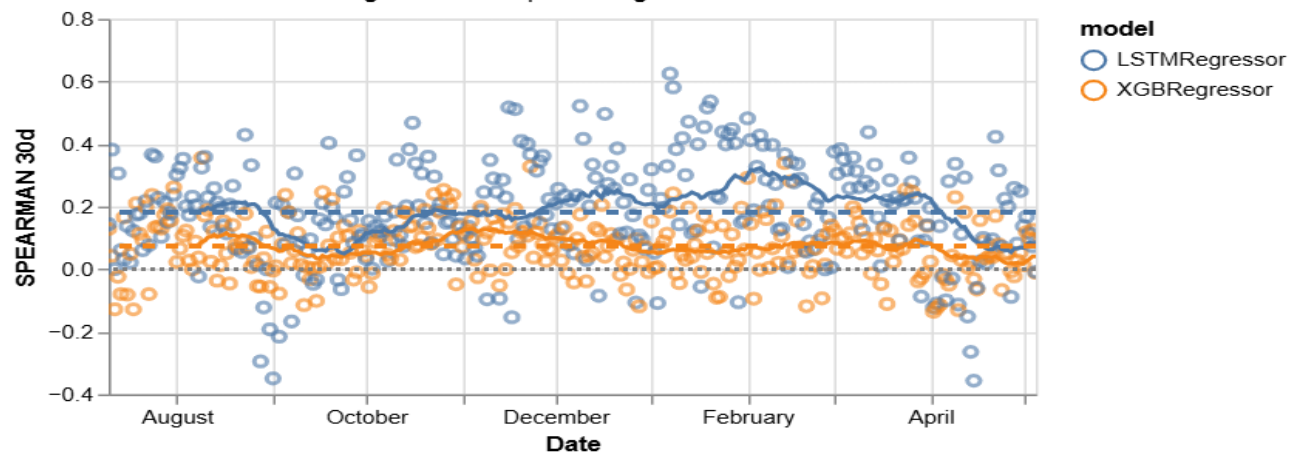
- What You Learn:
 - Proper time-series cross-validation with gap periods (TimeGapSplit).
 - Training & evaluating XGBoost vs LSTM fairly.
 - Using official scoring metrics & moving-average performance plots.
- Takeaway:
 - Sequential models like LSTM dramatically outperform tree models on CrowdCent's temporal crypto data — even without tuning.
- Where to find it:
 - This notebook is available here to run in colab: <https://docs.crowdcent.com/tutorials/advanced-cv-lstm/>

Tutorial Highlight: LSTM vs XGBoost on CrowdCent Data

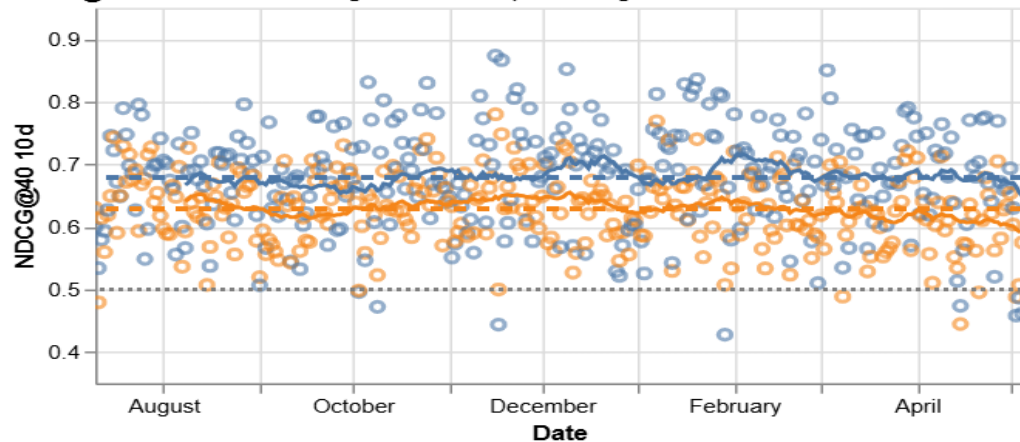
SPEARMAN 10d Means: XGBRegressor: 0.148 | LSTMRegressor: 0.255



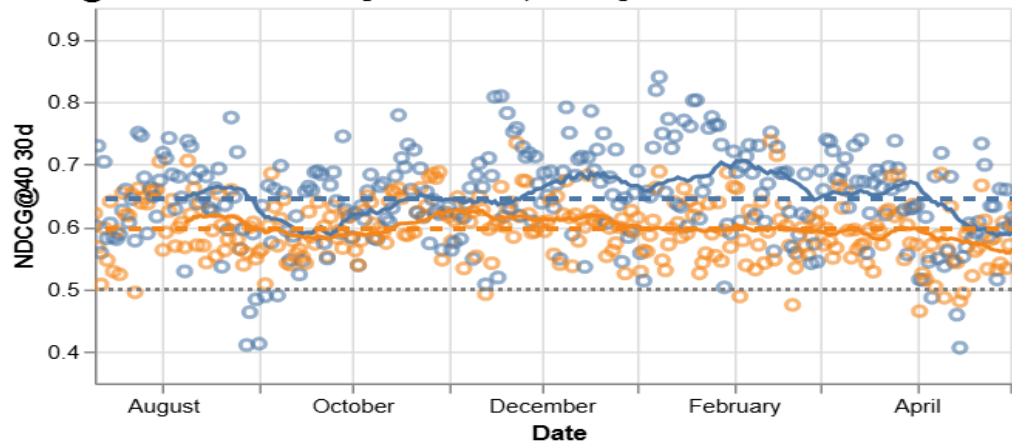
SPEARMAN 30d Means: XGBRegressor: 0.072 | LSTMRegressor: 0.182



NDCG@40 10d Means: XGBRegressor: 0.628 | LSTMRegressor: 0.678



NDCG@40 30d Means: LSTMRegressor: 0.645 | XGBRegressor: 0.597



Hyperliquid Ranking Meta Model

ⓘ Meta model signals are for informational purposes only. Not financial, investment, or trading advice. Use at your own risk.

RANK BY: PRED_30D ▾

🔍 BROWSE ALL ASSETS

📈 COMPARE ASSETS

🚀 FLIGHT SIMULATOR

📄 DOWNLOAD DATA

Quick jump to any asset (e.g., BTC, ETH, DOGE)...

🔍 GO

💡 Or browse all assets or compare multiple assets

Prediction Rankings as of Sep 05, 2025

⚠️ Model predictions are not guaranteed and subject to change

Strongest Positive Signals

#	Asset	Signal Strength
1	JUP	0.8598
2	ONDO	0.8573
3	ATOM	0.8236
4	ETH	0.7964
5	MNT	0.7894
6	BNB	0.7748
7	PENDLE	0.7735
8	POL	0.7658
9	DYM	0.7543
10	LTC	0.7486

Strongest Negative Signals

#	Asset	Signal Strength
159	VVV	0.2303
160	MOODENG	0.2223
161	PROVE	0.2103
162	SPX	0.1978
163	FXS	0.1915
164	GRIFFAIN	0.1842
165	MEME	0.1563
166	USUAL	0.1466
167	MAV	0.1405
168	BIO	0.1394

Asset Directory

Browse all 173 assets as of Sep 18, 2025

Filters & Search

Search

Min Signal

Max Signal

FILTER

RESET

Asset symbol...

0.0

1.0

Assets					
☐	Asset ↑	10d Signal	10d Rank	30d Signal	30d Rank
☐	AAVE	0.589	#65	0.487	#92
☐	ACE	0.303	#139	0.275	#148
☐	ADA	0.864	#6	0.838	#7
☐	AI16Z	0.639	#48	0.689	#35
☐	AIXBT	0.316	#137	0.182	#167
☐	ALGO	0.634	#50	0.629	#52
☐	ALT	0.606	#60	0.662	#38
☐	ANIME	0.366	#125	0.413	#121
☐	APE	0.713	#26	0.611	#57
☐	APT	0.862	#7	0.744	#18

ETH

[< ETC](#)[ETHFI >](#)

Asset prediction analysis from Jun 05, 2025 to Sep 05, 2025 (93 periods)

#60

10d Rank

#4

30d Rank

10d Signal: 0.579

30d Signal: 0.796

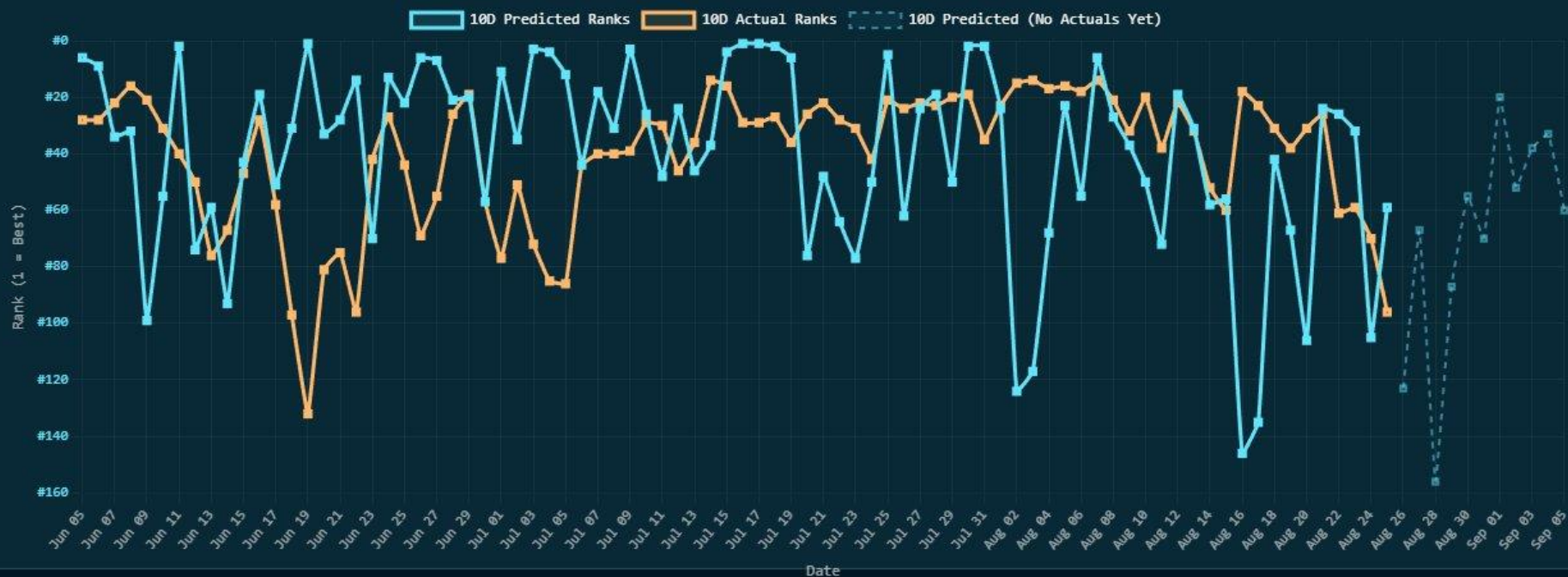
Best ranks: #1 / #1

Jun 2025 - Sep 2025 (93 periods)

Predictions vs Reality (10D)

[SIGNALS](#)[VS ACTUALS 10D](#)[VS ACTUALS 30D](#)[+ COMPARE](#)[BROWSE ALL](#)

Correlation: -0.16 Avg Error: 32.0 ranks Validated: 82



MCP Server

CROWDCENT

🚀 MCP Server for the Crowdcent Challenge

System Information:

- Platform: Linux 5.4.0-164-generic
- Python: 3.12.7
- Process ID: 2784985
- Started: 2025-06-23 16:30:37

🔧 Available Tools:

✓ list_all_challenges	✓ get_challenge_info
✓ switch_challenge	✓ list_training_datasets
✓ download_training_dataset	✓ download_inference_data
✓ submit_predictions_from_file	✓ submit_predictions_from_dataframe
✓ list_submissions	✓ get_submission
✓ download_meta_model	✓ get_training_dataset_info
✓ get_inference_data_info	

⚡ Server starting...

Build a model and submit to the crowdcent challenge

Generating...

Stop Ctrl+Shift+Q

I need to understand the task scope first. Let me gather information about the CrowdCent challenge and what's available.

> Called list_all_challenges ✓

> Called get_challenge_info ✓

Now let me get more details about the available training data and current inference data:

> Called list_training_datasets ✓

> Called get_inference_data_info ✓

Perfect! Now I understand the scope:

Task Clarification

Objective: Build and submit a machine learning model for the Hyperliquid Ranking challenge that predicts future relative price movements of crypto assets.

Challenge Details:

- Predict 10-day and 30-day forward relative returns (rankings)
- 80 features: 20 unique features × 4 lag values (0, 5, 10, 15 days)
- Current inference deadline: ~1.4 hours remaining
- Training data version 1.0 available

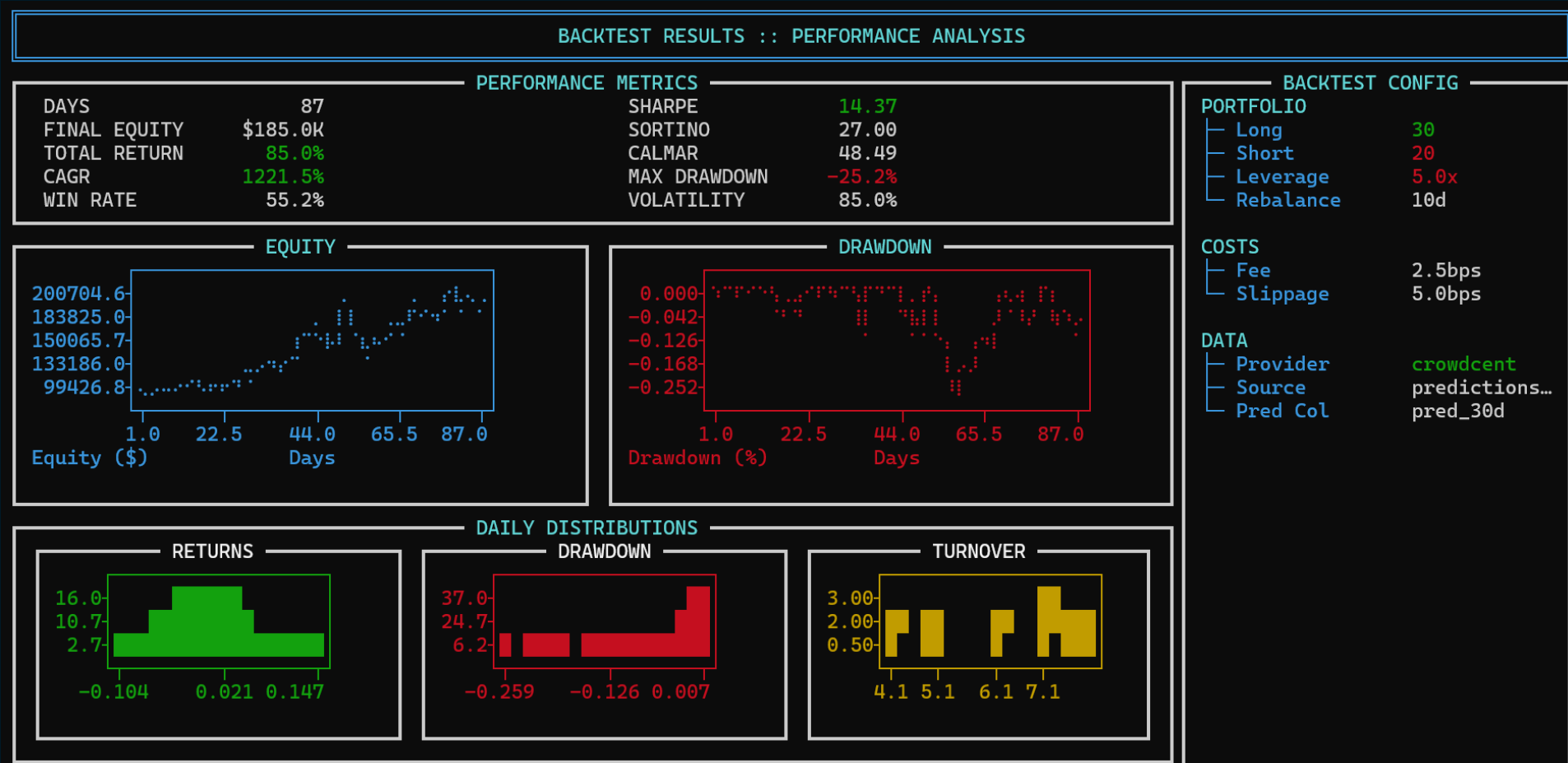
My Plan:

1. Download training data and current inference data
2. Build a simple but effective model (likely gradient boosting given time constraints)
3. Train on historical data with proper validation
4. Generate predictions for current inference period

Flight simulator



Creating portfolios and back-testing with cc-liquid



OPTIMIZATION RESULTS :: SHARPE

TOP 20 COMBINATIONS

Tested 375

Positive Sharpe 264/375 (70.4%)

RA...	LONG	SHORT	LEV	DAYS	SHARPE	CAGR	CALMAR	MAX DD	EQUITY
1	30	20	5.0x	10	14.37	1221.5%	48.49	-25.2%	\$185.0K
2	40	30	5.0x	10	13.06	872.9%	46.09	-18.9%	\$172.0K
3	30	20	5.0x	8	12.82	999.0%	30.29	-33.0%	\$177.1K
4	30	20	4.0x	10	10.80	734.7%	35.67	-20.6%	\$165.8K
5	40	30	4.0x	10	10.09	539.7%	35.05	-15.4%	\$155.6K
6	40	20	5.0x	10	10.05	1203.5%	35.04	-34.3%	\$184.4K
7	30	20	4.0x	8	9.87	615.2%	22.65	-27.2%	\$159.8K
8	50	30	5.0x	10	8.77	790.0%	28.65	-27.6%	\$168.4K
9	20	10	5.0x	8	8.13	992.3%	20.57	-48.2%	\$176.8K
10	30	20	3.0x	10	8.09	412.6%	26.13	-15.8%	\$147.6K
11	40	20	4.0x	10	8.08	774.2%	27.27	-28.4%	\$167.7K
12	40	30	3.0x	10	7.81	313.2%	26.68	-11.7%	\$140.2K
13	30	20	3.0x	8	7.57	354.0%	16.89	-21.0%	\$143.4K
14	50	40	5.0x	10	7.42	448.8%	26.04	-17.2%	\$150.1K
15	40	20	5.0x	8	7.40	872.1%	21.14	-41.2%	\$172.0K
16	50	30	4.0x	10	7.12	512.9%	22.66	-22.6%	\$154.1K
17	20	10	4.0x	8	6.78	662.9%	16.38	-40.5%	\$162.3K
18	20	20	5.0x	10	6.42	612.8%	24.82	-24.7%	\$159.7K
19	40	20	3.0x	10	6.32	453.9%	20.62	-22.0%	\$150.4K
20	50	40	4.0x	10	6.28	304.5%	21.95	-13.9%	\$139.5K

BEST BY SHARPE

BEST CONFIG

LONG 30
 SHORT 20
 LEVERAGE 5.0x
 REBALANCE 10 days

METRICS

SHARPE 14.37
 CAGR 1221.5%
 CALMAR 48.49
 MAX DD -25.2%
 VOLATILITY 85.0%

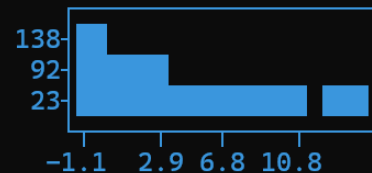
FINAL EQUITY \$185,016.66

DATA

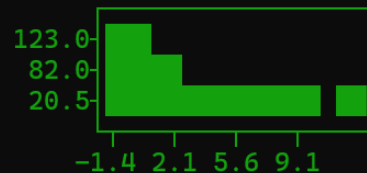
PROVIDER crowdcent
 SOURCE predictions...
 PRED COL pred_30d

METRIC DISTRIBUTIONS

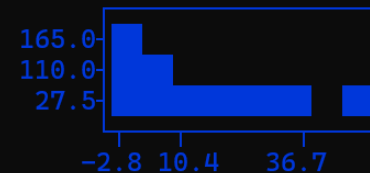
SHARPE



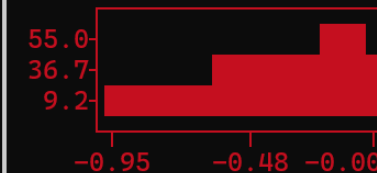
CAGR



CALMAR



MAX DD



CC-LIQUID MONITOR :: METAMODEL REBALANCER

METRICS

VALUE	\$100,000.00	LONG	\$240.0K		240%
MARGIN	\$33,333.33	SHORT	\$60.0K		60%
FREE	\$66,666.67	NET	\$180.0K	+180%	
LEVERAGE	3.00x	GROSS	\$300.0K	300%	

CONFIG

ENVIRONMENT

Network	TESTNET
Active Profile	default
Owner	0xYourMain
Vault	-
Signer Env	HYPERLIQUID_PRI...

DATA SOURCE

Provider	crowdcent
Path	predictions.par...
Prediction	pred_10d

PORTFOLIO

Long Positions	8
Short Positions	2
Target Leverage	3.0x
Rebalancing	
Frequency	Every 7 days
Time (UTC)	18:15

EXECUTION

Slippage	0.5%
Min Trade Value	\$10.00

POSITIONS

8L 2S | UNREALIZED \$+1,225.64 (+1.2%)

COIN	SIDE	SIZE	ENTRY	MARK	VALUE	PNL	PERF
BTC	L	0.50...	\$59.5K	\$60.0K	\$30.0K	+\$250.00	+0.8%
ETH	L	10.00	\$3.0K	\$3.0K	\$30.0K	+\$100.00	+0.3%
SOL	L	200.0...	\$149.00	\$150.00	\$30.0K	+\$200.00	+0.7%
AVAX	L	750.0...	\$39.50	\$40.00	\$30.0K	+\$375.00	+1.3%
ARB	L	24,0...	\$1.23	\$1.25	\$30.0K	+\$480.00	+1.6%
OP	L	15,7...	\$1.88	\$1.90	\$30.0K	+\$315.79	+1.1%
XRP	L	48,3...	\$0.61	\$0.62	\$30.0K	+\$483.87	+1.6%
MATIC	L	38,4...	\$0.79	\$0.78	\$30.0K	-\$384.62	-1.3%
LTC	S	384.0...	\$80.00	\$78.00	\$30.0K	+\$769.23	+2.5%
LINK	S	1,364	\$21.00	\$22.00	\$30.0K	-\$1,363.00	-4.8%

Next rebalance: 05:12:33

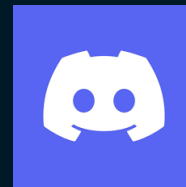
Last: 18.0h ago

Monitor refresh: 1.0s

Press Ctrl+C to exit

If you have any further queries, join us on Discord using
the link discord.gg/5BvpNSBsQ

Join



Check out the competition at
<https://crowdcent.com/challenge/hyperliquid-ranking/>

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